MATH 101 Final Exam - Practice 1

Version 202610122020.46 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Solve the nonlinear system algebraically.

$$x^2 + y^2 = 5$$
$$y = x + 1$$

2. Which pair satisfies h(x) = f(g(x))?

$$h(x) = (3x + 3)^2 + 4$$

A. $f(x) = x^2 + 4,\ g(x) = 3x + 3$

B. $f(x) = x^2 + 3,\ g(x) = 3x + 4$

C. $f(x) = 3x^2 + 4,\ g(x) = x + 3$

D. $f(x) = 3x + 3,\ g(x) = x^2 + 4$

3. Evaluate exactly using logarithmic properties; show the property used. Here log means base 10.

(*a*) $\log(0.25) + \log(400)$

(*b*) $\dfrac{\log_6(27)}{\log_6(9)}$

4. Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain.

$$f(x) = \log_2(x - 2) + 1$$

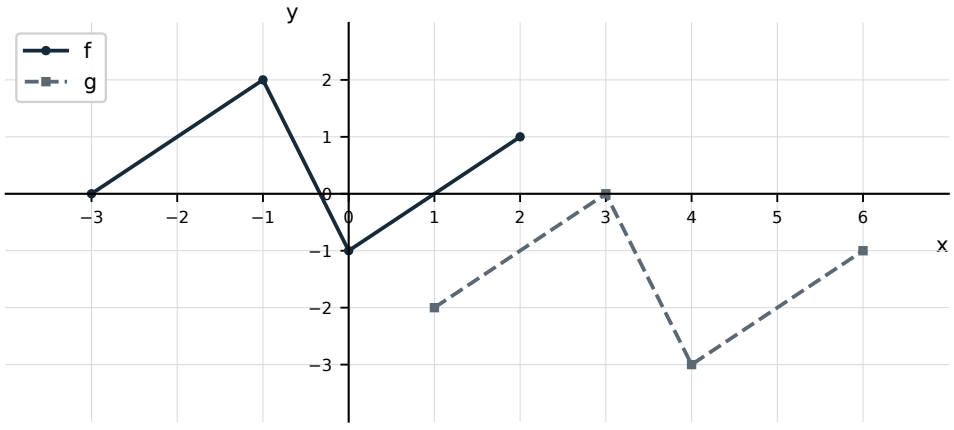
5. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x + 4) + 2$

B. $g(x) = f(x - 4) - 2$

C. $g(x) = f(x - 4) + 2$

D. $g(x) = f(x + 4) - 2$



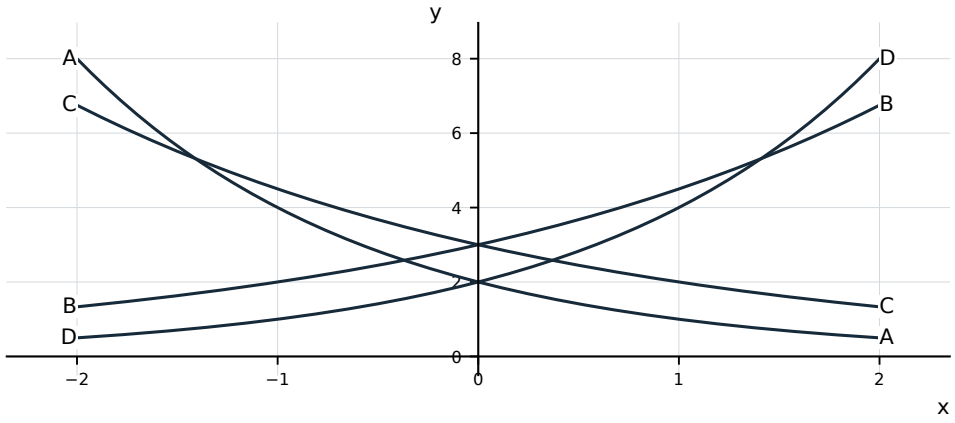
6. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 3(\frac{2}{3})^x$

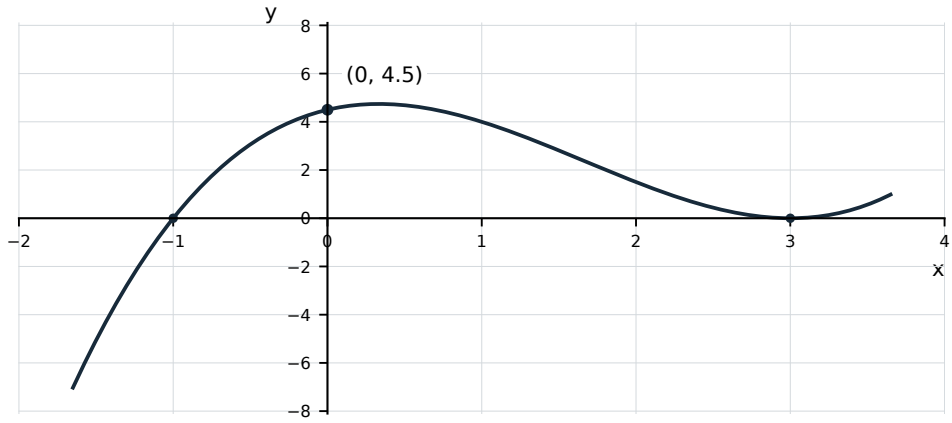
$(b)\quad 2(\frac{1}{2})^x$

$(c)\quad 2(2)^x$

$(d)\quad 3(\frac{3}{2})^x$



7. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



8. Find (a) domain and range; (b) both intercepts; (c) behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain.

$f(x) = \sqrt{x+4} - 2$

9. Find all zeros: (a) list Rational Root Theorem candidates, then test only 3 and -3; (b) use polynomial long division by the confirmed linear factor; (c) use the quadratic formula on the quotient. Record all zeros.

$p(x) = x^3 - 2x^2 - 4x + 3$

10. Find (a) domain; (b) hole, vertical asymptote, and horizontal asymptote; (c) both intercepts.

$f(x) = \frac{(x+2)(x+3)}{(x-5)(x+2)}$

Student answer sheet

MATH 101 Final Exam | Practice 1

Version 202610122020.46 | Name: _____

Question 1

Intersections	10 pts

Question 2

Choice	10 pts

Question 3

(a)	5 pts
(b)	5 pts

Question 4

(a) Domain	2 pts
(a) Range	2 pts
(b) Asymptote	1 pts
(b) x-intercept	1 pts
(c) Left end	2 pts
(c) Right end	2 pts

Question 5

Choice	10 pts

Question 6

(a)	3 pts
(b)	2 pts
(c)	3 pts
(d)	2 pts

Student answer sheet

MATH 101 Final Exam | Practice 1

Version 202610122020.46 | Name: _____

Question 7

Polynomial	10 pts

Question 8

(a) Domain	2 pts
(a) Range	2 pts
(b) x-intercept	1 pts
(b) y-intercept	1 pts
(c) Infinite end	3 pts
(c) Finite end	1 pts

Question 9

(a) Candidates/root	2 pts
(b) Quotient	4 pts
(c) Other zeros	4 pts

Question 10

(a) Domain	2 pts
(b) Hole	2 pts
(b) Vertical	1 pts
(b) Horizontal	1 pts
(c) x-intercept	2 pts
(c) y-intercept	2 pts

Answer key

MATH 101 Final Exam | Practice 1

Version 202610122020.46 | Name: _____

Question 1

Intersections	$(1,\ 2);\ (-2,\ -1)$	10 pts

Question 2

Choice	A	10 pts

Question 3

(a)	2	5 pts
(b)	$\frac{3}{2}$	5 pts

Question 4

(a) Domain	$(2, \infty)$	2 pts
(a) Range	$(-\infty, \infty)$	2 pts
(b) Asymptote	$x = 2$	1 pts
(b) x-intercept	$(\frac{5}{2},\ 0)$	1 pts
(c) Left end	$x \to 2^+ :\ f(x) \to -\infty$	2 pts
(c) Right end	$x \to \infty :\ f(x) \to \infty$	2 pts

Question 5

Choice	B	10 pts

Question 6

(a)	C	3 pts
(b)	A	2 pts
(c)	D	3 pts
(d)	B	2 pts

Answer key

MATH 101 Final Exam | Practice 1

Version 202610122020.46 | Name: _____

Question 7

Polynomial	$\frac{1}{2}(x+1)(x-3)^2$	10 pts

Question 8

(a) Domain	$[-4, \infty)$	2 pts
(a) Range	$[-2, \infty)$	2 pts
(b) x-intercept	$(0,\ 0)$	1 pts
(b) y-intercept	$(0,\ 0)$	1 pts
(c) Infinite end	$x \to \infty :\ f(x) \to \infty$	3 pts
(c) Finite end	$x \to -4^+ :\ f(x) \to -2$	1 pts

Question 9

(a) Candidates/root	$\{-3, -1, 1, 3\};\ x = 3$	2 pts
(b) Quotient	$x^2 + x - 1;\ R = 0$	4 pts
(c) Other zeros	$x = -\frac{1}{2} + \frac{\sqrt{5}}{2},\ \ -\frac{\sqrt{5}}{2} - \frac{1}{2}$	4 pts

Question 10

(a) Domain	$\mathbb{R} \setminus \{-2, 5\}$	2 pts
(b) Hole	$(-2,\ -\frac{1}{7})$	2 pts
(b) Vertical	$x = 5$	1 pts
(b) Horizontal	$y = 1$	1 pts
(c) x-intercept	$(-3,\ 0)$	2 pts
(c) y-intercept	$(0,\ -\frac{3}{5})$	2 pts